

Backend Models Review

Numbered issues to fix, one by one

This PDF is a checklist for your first backend project.
Every issue from the code review is listed in order.
Each item tells you what is wrong, why it matters,
which file to open, and the exact steps to fix it.

How this list is grouped

- Issues 1 to 5 Crash bugs. Fix these first.
- Issues 6 to 14 Same style everywhere.
- Issues 15 to 22 Model design (data rules).
- Issues 23 to 27 Missing pieces from the architecture.
- Issues 28 to 34 Habits to start before controllers.

How to use this PDF

You asked for a point-by-point list so you can fix things one at a time. That is exactly how this document is built. Do not try to fix everything in one sitting. Finish one number, tick it, then go to the next.

Suggested way to work:

- Print this PDF or keep it open beside Cursor.
- Start at Issue 1. Do not skip the crash bugs.
- After you change a file, save it and check that Node still starts.
- When an item is done, mark [] Done at the top of that issue.
- When Issues 1 to 27 are done, you are ready to write controllers.

Simple idea to remember

A model is a blueprint. It tells MongoDB what a document looks like. If the blueprint is wrong, every controller you write later will fight the database. Fixing models now is cheaper than rewriting controllers later.

What you are already doing well

This is your first backend, and several habits are already right. Keep these.

- You put allowed values in src/utils/constants.js instead of typing strings in every file.
- You write required: [true, "a clear message"] so users get a readable error.
- You use ref: "User" (and similar) instead of copying a whole user into every document.
- Most models use { timestamps: true }, so createdAt and updatedAt appear automatically.
- Your later models (BmisQueue, Payment, Notification) are cleaner than the early ones. That is normal. Copy the later style going forward.
- You already learned that type: Object skips nested field rules. That same lesson applies to type: Array.

Words used in this PDF (plain English)

Import

A file sayi

Export

A file sayi

Enum

A short lis

Ref / populate

A pointer
model na

Index

A lookup
duplicates

Snapshot

A frozen c
change of

Append-only

You may

Part A - Crash bugs (Issues 1 to 5)

These will throw errors as soon as the file is loaded, or `populate()` will fail. Fix this group before anything else. After each fix, run the server once to confirm it still starts.

1

ProgramCycle uses Scheme instead of Schema

FIX FIRST - this can crash or block the app ☐ Done

What is wrong

Why this matters (in simple words)

A little more explanation

Where to look

How to fix it

Near the t

JavaScript
is undefin
the whole

Think of it
asked for

src/model

1. Ope

2

ActivationProgress uses ENUMS but never imports it

FIX FIRST - this can crash or block the app ☐ Done

What is wrong

Why this matters (in simple words)

A little more explanation

Where to look

How to fix it

Activation
only impo

When No
Any route

You alrea
one line to

src/model

1. Add

3 SuccessProfile never exports the model

FIX FIRST - this can crash or block the app [] Done

What is wrong

The file c
is no exp

Why this matters (in simple words)

Other file
"#/models
save a pr

A little more explanation

Almost ev
is a small

Where to look

src/model

How to fix it

1. Scro

4 User looks up ENUMS.FOUNDER_TIER but constants names it founderTier

FIX FIRST - this can crash or block the app [] Done

What is wrong

In consta
(SCREAM

Why this matters (in simple words)

The found
may also
match.

A little more explanation

Your othe
founderTi
constants

Where to look

src/utlis/c

How to fix it

1. In c

5

FollowMeAllocation points at "Users" but the model is named "User"

FIX FIRST - this can crash or block the app [] Done

What is wrong

Why this matters (in simple words)

A little more explanation

Where to look

How to fix it

followerPa
model("Us

ref must n
"User", so

This is lik
wrong pla

src/model

1. Cha

Part B - Same style everywhere (Issues 6 to 14)

These will not always crash today. They will slow you down and create bugs that are hard to see. The goal is one pattern, copied in every model, so controllers feel boring and safe.

6

Two different ways of writing enums in models

CONSISTENCY - same pattern everywhere [] Done

What is wrong

Why this matters (in simple words)

A little more explanation

Where to look

How to fix it

Early mo
ENUMS.F
does not.

When you
do not. Th

Object.va
default wi

src/model

1.
ENUMS.N

7

User cannot be saved as a qualified Founder

CONSISTENCY - same pattern everywhere [] Done

What is wrong

Why this matters (in simple words)

A little more explanation

Where to look

How to fix it

constants
pending_
qualified,

The whole
official Fo
model is t

This happ
disappear

src/model
(FOUNDE

1. Eith

8

Spelling mistakes inside ENUMS that will leak into the API

CONSISTENCY - same pattern everywhere [] Done

What is wrong

Why this matters (in simple words)

A little more explanation

Where to look

How to fix it

Several o
"confrime
RECOMM

These str
checks fo
document

Fix spelli
Search th

src/utis/c
ACTIVAT

1. Cha

9

SuccessCenterCategory has "sucess" in names (missing one c)

CONSISTENCY - same pattern everywhere [] Done

What is wrong

The sche
The regis

Why this matters (in simple words)

The regis
controllers

A little more explanation

Same ide
model().

Where to look

src/model

How to fix it

1. Ren

10

Some models are missing timestamps

CONSISTENCY - same pattern everywhere [] Done

What is wrong

Most sch
AuditLog,
updatedA

Why this matters (in simple words)

Without ti
especially

A little more explanation

timestamp
that logic
the ledger

Where to look

src/model

How to fix it

1. Cha

11 Loose arrays and Mixed fields skip validation

CONSISTENCY - same pattern everywhere [] Done

What is wrong

Why this matters (in simple words)

A little more explanation

Where to look

How to fix it

programC
ContentB

You alrea
have the
.amount c

If you kno
[
status:
controller

src/model
src/model

1. Rep

12 LegalAcceptance imports ENUMS but does not use the context enum

CONSISTENCY - same pattern everywhere [] Done

What is wrong

Why this matters (in simple words)

A little more explanation

Where to look

How to fix it

You creat
The conte

People w
Reports a
never wire

Whenever
Otherwise

src/model
(LEGAL_

1. Set

3. Optional: default to `ENUMS.LEGAL_ACCEPTANCE_CONTEXT.SIGNUP`.

13**FounderPlan uses minlength on an array (that option is for strings)**

CONSISTENCY - same pattern everywhere [] Done

What is wrong

Why this matters (in simple words)

A little more explanation

Where to look

How to fix it

includedS
required"]

You think
Then Fou

For array:
validators

src/model

1. Ren

14**HTTP_STATUS is missing 409, and error middleware uses a raw 409**

CONSISTENCY - same pattern everywhere [] Done

What is wrong

Why this matters (in simple words)

A little more explanation

Where to look

How to fix it

constants
409 and s

The whole
forget wh
habit: nev

409 mean
and use th

src/utl/c

1. Add

Part C - Model design (Issues 15 to 22)

These are about how data is stored and protected. If you skip them, controllers will create duplicate rows, wrong money, and Founders whose benefits change when Todd edits a plan.

15

Add compound unique indexes for "one row per user + program"

MODEL DESIGN - fix before controllers [] Done

What is wrong

You index
Business
database.

Why this matters (in simple words)

Without a
same use
You will s

A little more explanation

A compou
can selec
In Mongo
partial ind
+ program

Where to look

SuccessC
legalDocu

How to fix it

1. For
(use parti

16

Do not store money as a normal JavaScript Number forever

MODEL DESIGN - fix before controllers [] Done

What is wrong

amount, g
numbers

Why this matters (in simple words)

This is a r
you will ne

A little more explanation

Two com
when you
Cents is s
be exact c

Where to look

Every mo
Activation
SuccessP

How to fix it

1. Dec

17

FounderPurchase is missing snapshots of price and benefits

MODEL DESIGN - fix before controllers [] Done

What is wrong

FounderP
name, pri
founderBe

Why this matters (in simple words)

Todd will
bought. If
may have

A little more explanation

A snapsh
benefitsV
PDF listed

Where to look

src/model

How to fix it

1. Add

18

Payment and FounderPurchase point at each other and both want to be required

MODEL DESIGN - fix before controllers [] Done

What is wrong

Payment.
create Pa

Why this matters (in simple words)

This is a
"create en

A little more explanation

Pick one
pending.
completed
transactio

Where to look

src/models/Paymen

How to fix it

1. Kee

19

Auto-increment order on Category and Program can clash

MODEL DESIGN - fix before controllers [] Done

What is wrong

A pre("sa
both read

Why this matters (in simple words)

You will g
condition:

A little more explanation

For Phas
order from
Also, orde
order that

Where to look

src/model

How to fix it

1. For

20

Password hashing only runs on .save(), not on findByIdAndUpdate

MODEL DESIGN - fix before controllers [] Done

What is wrong

userSche
User.findE

Why this matters (in simple words)

If a futur
password

A little more explanation

Mongoos
It also ru
MongoDB
newPassv

Where to look

src/model

How to fix it

controller.

1. Kee

21

fullName virtual will not appear in API JSON

MODEL DESIGN - fix before controllers [] Done

What is wrong

You added
include vir

Why this matters (in simple words)

You will t
surprise.

A little more explanation

Enable it
password

Where to look

src/model

How to fix it

1. Cha

22

User.status defaults to active while emailVerified defaults to false

MODEL DESIGN - fix before controllers [] Done

What is wrong

A new us
verified bo

Why this matters (in simple words)

Controller
might buy

A little more explanation

Make the
and email
AUTH_ST
extra stat
status" fro

Where to look

src/model

How to fix it

1. Set status default to `ENUMS.AUTH_STATUS.PENDING_VERIFICATION`.

2. Keep emailVe

Part D - Architecture gaps (Issues 23 to 27)

These match Todd's architecture PDF. They are not typos. They are missing pieces. If you start controllers without them, business rules will be hidden inside Express handlers and will be hard to test.

23 founderEntitlements collection is missing

MODEL DESIGN - fix before controllers [] Done

What is wrong

The archi
and Four
FounderP

Why this matters (in simple words)

Without th
That cann
program.

A little more explanation

One purc
Each row
eligibility.
document

Where to look

New file: s

How to fix it

1. Add
ineligible)

24 FinancialLedger should be append-only (no quiet edits)

MODEL DESIGN - fix before controllers [] Done

What is wrong

The archi
updatedA

Why this matters (in simple words)

If a contro
reversal r

A little more explanation

Two practical steps to fix this are: update the `updateOne` method to use `updateOne` instead of `update`. You already have `updateOne` in the `pre("updateOne")` hook, so you can just call `updateOne` instead of `update`.

Where to look

`src/models`

How to fix it

1. Replace

25

Follow Me 12.5 percent defaults are hard-coded

MODEL DESIGN - fix before controllers [] Done

What is wrong

directedAt
is: all rule

Why this matters (in simple words)

If Todd changes the
get 12.5 percent
(snapshot)

A little more explanation

The model has
settings (for
does not

Where to look

`src/models`

How to fix it

1. Remove

26

Role naming: architecture says participant, code says user

CONSISTENCY - same pattern everywhere [] Done

What is wrong

The architecture says
The User

Why this matters (in simple words)

This is not a good
for "participant"

A little more explanation

"user" is a

Where to look

How to fix it

real data

src/utls/c

1. Dec

27

Clean leftover comments and commented-out rules

CONSISTENCY - same pattern everywhere [] Done

What is wrong

Several n
futureProg

Why this matters (in simple words)

Comment
dangerou
FounderP

A little more explanation

If a field is
That noise

Where to look

SuccessC

How to fix it

1. Dele

Part E - Habits before you write controllers (Issues 28 to 34)

You do not need to build all of this on day one of controllers. You should decide the shape now, so the first route does not invent a second style.

28

Add a small AppError helper

BACKEND HABIT - start this before controllers [] Done

What is wrong

errorMidd
res.status

Why this matters (in simple words)

If every c
and the fr

A little more explanation

Create a
controller:

Where to look

How to fix it

false, mes

New file ic

1. Cre

29

Do not send raw err.message on unexpected 500 errors

BACKEND HABIT - start this before controllers [] Done

What is wrong

errorMidd
can be a l

Why this matters (in simple words)

Attackers
should sta

A little more explanation

Pattern: if
log the res

Where to look

src/middle

How to fix it

1. Kee

30

.env.example is gitignored, so nobody sees which env vars are needed

BACKEND HABIT - start this before controllers [] Done

What is wrong

.gitignore
keys: MO

Why this matters (in simple words)

The next
process.e

A little more explanation

Commit .e

Where to look

.gitignore

How to fix it

1. Ren

31 Agree on folders before the first controller

BACKEND HABIT - start this before controllers [] Done

What is wrong

Right now
appear, p

Why this matters (in simple words)

If BMIS ru
test) with
applies th

A little more explanation

A simple
Services
uses two

Where to look

Plan thes

How to fix it

1. Cre

32 Validate req.body at the door, not only in Mongoose

BACKEND HABIT - start this before controllers [] Done

What is wrong

Mongoose
still pass

Why this matters (in simple words)

You alrea
password
before yo

A little more explanation

This also
to user, v
safer.

Where to look

Future: sn

How to fix it

1. Whe

33

Use a MongoDB transaction for Founder purchase (money + entitlements)

BACKEND HABIT - start this before controllers [] Done

What is wrong

Founder
permanen

Why this matters (in simple words)

Without a
Support c
entitlemen

A little more explanation

MongoDB
const ses
this on He
three sep

Where to look

Future Fo

How to fix it

1. Do n

34

Enable virtuals / id and a consistent API response shape

BACKEND HABIT - start this before controllers [] Done

What is wrong

Your heal
forever. W
sometime

Why this matters (in simple words)

Frontends
Mongoos

A little more explanation

This is no

Where to look

src/app.js

How to fix it

1. Suc

Suggested order of work this week

If you only have a few hours, do not start at Issue 34. Use this order. Each group should leave the project in a working state.

Day 1 - Make the models load

- Issues 1, 2, 3, 4, 5. Server should start. Imports should work.
- Quick test: import every model from a small script, or just boot nodemon and watch for red errors.

Day 2 - One enum style and spelling

- Issues 6, 7, 8, 9, 12, 14. constants.js and User/SuccessProfile enums.
- Search the repo for the old misspellings after you rename.

Day 3 - Timestamps, arrays, indexes, purchase shape

- Issues 10, 11, 13, 15, 17, 18.
- Add FounderEntitlement (Issue 23) on this day if you can. Purchase depends on it.

Day 4 - Money, ledger, defaults

- Issues 16, 22, 24, 25, 26.
- Write down the cents decision somewhere you will see it.

Day 5 - Hygiene and controller prep

- Issues 19, 20, 21, 27, 28, 29, 30, 31, 32, 33, 34.
- Create folders. Create AppError. Create .env.example.
- Then write only signup / login. Nothing else.

When are you "done" with models?

When Issues 1 to 5 are gone, every model exports, enums match constants, purchase can snapshot a plan, entitlements exist, and you have unique indexes for user+program. You do not need perfect money-cents on every field before signup. You do need it before Payment goes live.

First controllers (after this list)

Do not start with Avalanche or BMIS queues. This order will teach you the stack without money risk:

- 1. Auth: signup, login, me, verify email.
- 2. Users: read/update own profile (no password via findByIdAndUpdate).
- 3. Success Profile: create/update the one profile per user (you already unique-index userId).
- 4. Catalog: list categories and programs (read-only for participants).
- 5. Founder plans + purchase (transactions, snapshots, entitlements).
- 6. Success Center selection + recommendations.
- 7. Activation progress, program cycles, then queues and ledger (simulation).

Quick tick list (print this page)

[] 1 ProgramCycle Schema typo

[] 2 Ac

You are doing real backend work: models, enums, refs, and an architecture that separates money from display. The list above is not a failure report. It is a map. Tick it in order, then write auth. If you want help applying a group of issues in the repo next, start with Part A (1 to 5) in Cursor.